1. Statements a, d, f, g, and i are equivalent, and statements b, c, e, and h are equivalent. □

2. Consider a quadrilateral ABCD. Let's say statement P is that ABCD is a trapezoid, and Q is that the diagonal angles in ABCD each sum to 180. Then P implies Q, but Q does not imply P. □

3. We can rewrite the inverse and converse as such:
   Inverse: if P is false, Q is false
   Converse: if Q is true, P is true

   First, let's assume the inverse is true. That means Q is false whenever P is false. The converse has the condition that Q is true. Since the inverse says that Q is false when P is false, P must be true, because Q isn't false. Therefore, assuming that the inverse is true, we can come to the conclusion that when Q is true, P is true, which is exactly what the converse states.

   Now, let's assume the inverse is false. That means if P is false, Q isn't necessarily false. So if Q is true, there is a possibility that P is false. That contradicts the converse, which states that if Q is true, P is true. Therefore, if the inverse is false, the converse is false. □

4. The first statement says that for all x, there exists y such that P(x, y), and the second states that there exists y for all x such that P(x, y). If we assume the first one is true, we know that we can find y such that P(x, y), no matter what x is. That means the second statement is also true.

   If we assume the first statement is false, then for all x, there doesn't exist y such that P(x, y). That means the second statement is false, because it states that there actually exists y for all x such that P(x, y). Because the second statement is true when the first statement is true and the second statement is false when the first statement is false, both statements are equivalent. □

5. Proofs:
   a. We can factor $x^2 + 5x$ into $x(x + 5)$. From there, there are two cases. The first case is that x is even. If so, then multiplying x by $x + 5$ would result in an even number, so $x^2 + 5x$ is even. If x is odd, $x + 5$ is even, so multiplying the two would result in an even number, regardless of whether x is even or odd. □

   b. When you raise an odd number to a power, the result is also odd because it has no power of 2 in its prime factorization. So $x^3 - 3x^2 + 5x + 7$ is the same thing as combining four odd numbers, which results in an even number. □

c. If $x^2 + 2x + 3$ is even, then $x^2 + 2x$ is odd. And since $2x$ is even, x^2 must be odd to make their sum odd. Therefore, x must be odd. □

d. If n is even, then n^2 is a multiple of 4. If n is odd, then the product of the number one more than n and one less than n is a multiple of 4, because they are both even. So $(n - 1)(n + 1)$ is a multiple of 4. Expanding that, we get $n^2 - 1$. So if n is an integer, either n^2 or $n^2 - 1$ is a multiple of 4. □

6. First of all, $x(x + 1)(x + 2)(x + 3)$ is a product of four consecutive numbers. That means it has two factors of 2, of which one is a factor of 4. That means the product is divisible by 8. Also, it must have at least one multiple of 3, so the whole product is divisible by 24.

   Beyond that, there is no guarantee that the product is divisible by any more prime factors. Setting $x = 1$, we get the product of 24. That can't be divisible by anything greater than 24. Therefore, 24 is the greatest possible divisor of $x(x + 1)(x + 2)(x + 3)$. □

7. Proofs:
   a. First, let's assume $\sqrt{3}$ is rational by setting it equal to a/b for relatively prime integers a and b. Squaring both sides and then multiplying by b^2, we get $a^2 = 3b^2$. Since b is an integer, a has to be a multiple of 3, so we can say $a = 3c$ for some integer c. Substituting that into $a^2 = 3b^2$ and simplifying further, we can arrive at $b^2 = 3c^2$. Since c is an integer, b must be a multiple of 3. But that contradicts our original statement that a and b were relatively prime, since they cannot both be a multiple of 3. Therefore, our assumption that $\sqrt{3}$ is rational is incorrect. □

   b. First, let's assume $\sqrt[3]{4}$ is rational by setting it equal to a/b for relatively prime integers a and b. Cubing both sides and then multiplying by b^3, we get $a^3 = 4b^3$. Since b is an integer, a has to be a multiple of 2 for a^3 to be a multiple of 4, so we can say $a = 2c$ for some integer c. Substituting that into $a^3 = 4b^3$ and simplifying further, we can arrive at $b^3 = 4c^3$. Since c is an integer, b must be a multiple of 2. But that contradicts our original statement that a and b were relatively prime, since they cannot both be a multiple of 2. Therefore, our assumption that $\sqrt[3]{4}$ is rational is incorrect. □

   c. While following the same steps we did in the previous proofs, we arrive at the equation $9b^2 = a^2$, which means $a = 3c$ for some integer c. However, substituting that and simplifying, we arrive at $b^2 = c^2$, which does not tell us that b is also a multiple of 3. Therefore, this proof has failed, and we cannot come to the conclusion that $\sqrt{9}$ is irrational. □

8. Proofs:
   a. Let's say x is the irrational number and y is the rational number. That means we can write y as a/b, where a and b are integers. To prove this by contradiction, let's say $x + y$ is rational, or $x + a/b = c/d$ for integers c/d. We can subtract a/b

from both sides to get $x = c/d - a/b$, and then we can combine both fractions to get $x = (bc - ad)/bd$. But since $bc - ad$ and bd are both rational, the equation states that x is rational, which is a contradiction. Therefore, the sum of an irrational number and a rational number is irrational. □

b. The sum of the irrational numbers, $\sqrt{2}$ and $-\sqrt{2}$, is 0, which is rational. □

9. We can prove this by contradiction. Let's say that the irrational number is x and the rational number is y. Since y is rational, we can write it as a/b where a and b are integers. If we assume that xy is rational, we can say $xy = c/d$. Substituting y with a/b, we get $x = bc/ad$, which is an integer over another integer. That means x is rational, but that contradicts our original statement that x is rational. Therefore, our assumption that the product is rational is wrong, and xy is irrational. □

10. We can prove this by contradiction. First, we assume that $\log_2(3)$ is rational, so it can be written in the form a/b, where a and b are integers. From there we can say $3 = 2^{a/b}$, and then $3^b = 2^a$. However, that is impossible, because the number on the left would be a power of 3, and the one on the right would be a power of 2. That means they cannot be equal. Therefore, our assumption that $\log_2(3)$ is rational is false, so it's irrational. □

11. First, let's say $x = \sqrt{2}^{\sqrt{2}}$, and $y = \sqrt{2}$. We are given that $x^y = 2$, which is rational. We can rewrite x as $2^{\sqrt{2}/2} = \sqrt{(2^{\sqrt{2}})}$, which is irrational. Therefore, both x and y are irrational and x^y is rational. □

12. Let's consider four cases:

    Case 1: x and y are even.
    If both x and y are even, then we can say $x = 2m$ and $y = 2n$ for integers m and n. Substituting that into $x^2 = 3y^2 + 2$, expanding, and then dividing by 2, we get $2m^2 = 1 + 6n^2$. The left side is even because it is twice an integer. However, the right side is odd because it is one more than twice an integer. Therefore, we have a contradiction, and there are no even x and y which satisfy the equation $x^2 - 3y^2 = 2$.

    Case 2: x is even, y is odd.
    In this case, x^2 is even and y^2 is odd, which means $3y^2$ is odd, and the difference between the two is also odd. Therefore, it can't be equal to 2.

    Case 3: x is odd, y is even.
    In this case, x^2 is odd and y^2 is even, which means $3y^2$ is even, and the difference between an odd number and an even number is odd, so it can't be 2.

    Case 4: x and y are odd.
    If x and y are odd, $x = 2m + 1$ and $y = 2n + 1$ for integers m and n. We can plug those into the equation $x^2 = 3y^2 + 2$ to get $(2m + 1)^2 = 3(2n + 1)^2 + 2$. Expanding that, we get

$4m^2 + 4m + 1 = 12n^2 + 12n + 5$. Then, rearranging that and dividing by 4, we get $m^2 + m - 3n^2 - 3n - 1 = 0$. We can rewrite that as $m(m + 1) - 3n(n - 1) = 1$. We know that $m(m + 1)$ and $n(n - 1)$ must both be even because they are the products of two consecutive integers. So the left side of the equation must be even. However, the right side is 1. Therefore, there is a contradiction, and so there are no solutions to $x^2 - 3y^2 = 2$ where x and y are both odd.

We considered all possible cases of x and y, and proved that in each case there were no solutions. Therefore, there are no integers x and y such that $x^2 - 3y^2 = 2$. □